

Lab 02: Elementary Prog. II

Objective

The purpose of this Lab. is to familiarize students how to solve practical problems programmatically; they will practice on elementary programming using primitive data types, variables, constants, operators, expressions, and input and output. Also, they will learn how to diagnose errors that may occur when a program is compiled or executed. There are some exercises, through which they will understand the concepts they learnt in this chapter.

Current Lab Learning Outcomes (LLO)

By completion of the lab the students should be able to

Understand an enhanced focus on using ideas and methods, learned in class, to help them solve real world problems

Lab Requirements

NetBeans

Lab Description

Reading Input from the Console

The program can be better, and more interactive, by letting the user enter values of variables from the keyboard. The following statements are used to input the values from the keyboard.

- `Scanner scanner_name= new Scanner(System.in);`
- `System.out.print("Any Suitable Message: ");`
- `var_name= scanner_name.input_method;`
- The character with red color should be capital letter.
- `scanner_name` : any identifier name
- **Any Suitable Message**: any understandable message for user to enter the value from the keyboard (for example "Enter the name: " for entering the name **or** "Enter the Radius of Circle: " to enter the value of radius and so on).
- **input_method**:
 - `next()`: if the `var_name` is of type `String`
 - `nextInt()`: if the `var_name` is of type `int`
 - `nextDouble()`: if the `var_name` is of type `double`
 - `nextFloat()`: if the `var_name` is of type `float`
 - `nextByte()`: if the `var_name` is of type `byte`
- `Scanner scanner_name= new Scanner(System.in);` is written only one time in the program. Before using this statement, we need to write the statement `import java.util.Scanner;` before the class name and at the beginning of file.

Practice Activity with Lab Instructor (10 minutes)

(Physics: acceleration) Average acceleration is defined as the change of velocity divided by the time taken to make the change, as shown in the following formula:

$$a = \frac{v_1 - v_0}{t}$$

Write a program that prompts the user to enter the starting velocity v_0 in meters/ second, the ending velocity v_1 in meters/second, and the time span t in seconds, and displays the average acceleration. Here is a sample run:

Enter v0, v1, and t: 5.5 50.9 4.5
The average acceleration is 10.0889

Input: Value of v_0, v_1 and t .

Processing: Apply the above equation to find the average acceleration (a).

Output: Display the average acceleration (a) same to the output format above.

Solution:

1. Open NetBeans and create a new project
2. Create a new java main class and write its name as AverageAcceleration
3. Write the following code inside the main method

```
Source History
1 import java.util.Scanner;
2 public class AverageAcceleration {
3     public static void main(String[] args) {
4         Scanner input = new Scanner(System.in);
5         // Input values of v0, v1 and t
6         System.out.print("Enetr v0, v1 and t: ");
7         double v0 = input.nextDouble();
8         double v1 = input.nextDouble();
9         double t = input.nextDouble();
10
11         //compute the average acceleration
12         double aveAcce = (v1 - v0) / t;
13
14         // display the average acceleration
15         System.out.println("The average acceleration is " + aveAcce);
16     }
17 }
```

Output - LabManualsprorams (run) x

run:
Enetr v0, v1 and t: 5.5 50.9 4.5
The average acceleration is 10.088888888888889
BUILD SUCCESSFUL (total time: 22 seconds)

- Short Exercise (5 minutes)

Do the above program and run it many times with different values.

Lab Assessment

Three Problems\ exercises should be completed **in 60 minutes individually**

1. **(Physics: finding runway length)** Given an airplane's acceleration a and take-off speed v , you can compute the minimum runway length needed for an airplane to take off using the following formula:

$$\text{length} = \frac{v^2}{2a}$$

Write a program that prompts the user to enter v in meters/second (m/s) and the acceleration a in meters/second squared (m/s²), and displays the minimum runway length. Here is a sample run:

```
Enter speed and acceleration: 60 3.5 Enter
The minimum runway length for this airplane is 514.286
```

2. **(Sum the digits in an integer)** Write a program that reads an integer between 0 and 1000 and adds all the digits in the integer. For example, if an integer is 932, the sum of all its digits is 14. Hint: Use the % operator to extract digits, and use the / operator to remove the extracted digit. For instance, $932 \% 10 = 2$ and $932 / 10 = 93$. Here is a sample run:

```
Enter a number between 0 and 1000: 999 Enter
The sum of the digits is 27
```

3. **(Cost of driving)** Write a program that prompts the user to enter the distance to drive, the fuel efficiency of the car in miles per gallon, and the price per gallon, and displays the cost of the trip. Here is a sample run:

```
Enter the driving distance: 900.5 Enter
Enter miles per gallon: 25.5 Enter
Enter price per gallon: 3.55 Enter
The cost of driving is $125.36
```